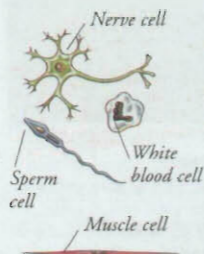


CELLS



ALL LIVING ORGANISMS are made of self-contained units of life called cells. Some, such as the amoeba, consist of a single cell, while others, such as humans, are made up of billions of cells. Each cell has a nucleus that contains the genetic material DNA, which provides the instructions the cell needs to maintain itself. Surrounding the nucleus is the cytoplasm, which contains the matter that makes the cell function. Forming a layer around the cytoplasm is the cell membrane, which forms the cell's boundary.



Specialized cells

Most plants and animals consist of many cell types, each specialized to perform a specific task. Neurones are long cells that carry nerve impulses around an animal's body; guard cells are rigid box-like structures filled with fluid. They open and close pores on the surface of plant leaves.



Palisade mesophyll cell

These cells are found in the upper layer of the middle part, or mesophyll, of plant leaves. They are packed with chloroplasts, which contain the green pigment chlorophyll that harnesses the energy in sunlight.

Palisade mesophyll cell



Liver cells

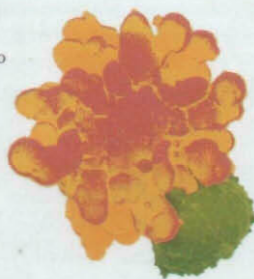
The human liver has over 500 functions related to controlling the chemical balance of the body. These functions are carried out by cells called hepatocytes. For instance, some liver cells remove poisons from blood.

Liver cell

Abnormal cells

When cells divide inside an organism they do so in a controlled way. Sometimes, cells become abnormal and start dividing uncontrollably, leading to the production of growths called tumours. The presence of these abnormal cells and tumours causes a number of different forms of a disease called cancer.

Cancer tumour cell (yellow) being attacked by a T-lymphocyte cell (green).



Marie-François Bichat

French pathologist Marie-François Bichat (1771–1802) showed that an organ, whether a leaf of a plant or a kidney of an animal, is made of different groups of cells. He called each group a tissue, and showed that the same tissues could appear in different organs. His research formed the basis of histology – the study of organs and tissues.



Model of an animal cell

Endoplasmic reticulum is a maze-like network of membranes that make and store chemicals.

Golgi apparatus sorts and stores proteins.

Nucleus is the cell's control centre.

Vacuole is a small and temporary space where food and waste is stored.

Glycogen granules are food reserves or insoluble waste.

Pinocyte allows substances to filter in and out of cell.

Cell structure

Most cells have similar structures. They consist of a fluid called cytoplasm, a surrounding cell membrane, and a nucleus. Cytoplasm contains structures known as organelles. Plant cells, unlike animal cells, have a tough outer wall and chloroplasts.

Plasma membrane is the thin flexible layer surrounding the cytoplasm.

Mitochondrion generates energy from sugars and fatty acids.

Cytoplasm forms the bulk of the cell and gives it its shape.

Organelles are any structures that live in the cytoplasm and control special functions.

Nucleolus, centre of nucleus

Cytoplasm

Model of a plant cell

Cellulose cell wall is a tough outer jacket mainly made of cellulose.

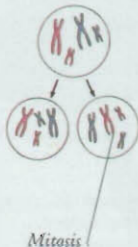
Plasma membrane is selectively permeable or semi-permeable and is concerned with receiving stimuli.

Chloroplast is an organelle present in green plants; it converts light energy into food by photosynthesis.

Vacuole is a clear space filled with fluid.

Cell division

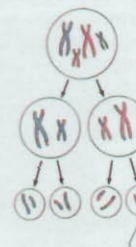
Cells reproduce by dividing. During cell division the nucleus divides first, followed by the cytoplasm. There are two kinds of cell division: mitosis and meiosis. Mitosis produces cells needed for growth and to replace dead cells. Meiosis produces sex cells for reproduction.



Mitosis

Mitosis

This produces two daughter cells that are identical to the parent cell. The cell's chromosomes (genetic material) make copies of themselves. These separate and move to opposite ends of the cell to form two new nuclei. The cytoplasm splits and two new cells are formed.



Meiosis

Meiosis

This takes place in sex organs and involves two cell divisions. It makes four sex cells that differ from the parent cells because they have half the normal number of chromosomes. These sex cells are called sperm in males and ova in females.

Studying cells

Cells are so small they need to be studied with a microscope. Both the light microscope and the electron microscope have revealed cells' external and internal structure. For this study cells must be carefully prepared to see their details clearly.



Chemical dyes used for staining cells

Staining cells

When cells are seen under a microscope they are often transparent, showing little detail. For that reason, they are coloured with chemical stains to pick out details such as the nucleus.